SQL Operators

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Example Table: STUDENT

- STUDENT(SID, Name, DeptID, Marks)

SID	Name	DeptID	Marks
1	Amit	10	85
2	Riya	20	72
3	Rohit	10	60
4	Neha	30	45
5	Arjun	20	NULL

Arithmetic Operators (+, -, *, /)

Example: Add 5 marks to each student

Query:

- SELECT Name, Marks + 5 AS **NewMarks** FROM STUDENT;

Output:

Name	NewMarks
Amit	90
Riya	77
Rohit	65
Neha	50
Arjun	NULL

Comparison Operators (=, >, <, <>)

Example: Students scoring > 70

Query:

- SELECT Name, Marks FROM STUDENT WHERE Marks > 70;

Output:

Name	Marks
Amit	85
Riya	72

Example: Students NOT in DeptID 10

```
SELECT Name, DeptID
FROM STUDENT
WHERE DeptID <> 10;
```

Name	DeptID
Riya	20
Neha	30
Arjun	20

Logical Operators (AND, OR, NOT)

- Example: DeptID=10 AND Marks>70

- Query:

SELECT Name FROM STUDENT WHERE DeptID=10 **AND** Marks>70;

- Output:

Name	Marks
Amit	85

Example: DeptID = 10 OR DeptID = 30

```
SELECT Name, DeptID
FROM STUDENT
WHERE DeptID = 10 OR DeptID = 30;
```

Name	DeptID
Amit	10
Rohit	10
Neha	30

NOT Example: NOT DeptID = 20

```
SELECT Name, DeptID
FROM STUDENT
WHERE NOT DeptID = 20;
```

Name	DeptID
Amit	10
Rohit	10
Neha	30

Special Operator: IN

Example: Students from DeptID 10 or 20

Query:

- **SELECT Name FROM STUDENT WHERE DeptID IN (10,20);**

- Output:

Name	DeptID
Amit	10
Riya	20
Rohit	10
Arjun	20

Special Operator: BETWEEN

- Example: Marks between 60 and 80

Query:

- SELECT Name FROM STUDENT WHERE Marks BETWEEN 60 AND 80;

- Output:

Name	Marks
Riya	72
Rohit	60

Special Operator: LIKE

- Example: Names starting with 'R'

- Query:
- SELECT Name FROM STUDENT WHERE Name LIKE 'R%';

- Output:
- Riya, Rohit

Name
Riya
Rohit

Special Operator: IS NULL

- Example: Students with NULL marks

Query:

- SELECT Name FROM STUDENT WHERE Marks IS NULL;

Name	Marks
Arjun	NULL

- Output:
- Arjun

Set Operators (UNION, INTERSECT, EXCEPT)

Example:

- UNION combines results of two queries.
- INTERSECT gives common rows.
- EXCEPT gives difference.

(Used in advanced queries)

ENROLLMENT Table

SID
1
2
6

STUDENT Table

SID	Name	DeptID	Marks
1	Amit	10	85
2	Riya	20	72
3	Rohit	10	60
4	Neha	30	45
5	Arjun	20	NULL

UNION Example

- **Combine student IDs from both tables**

SELECT SID FROM STUDENT

UNION

SELECT SID FROM ENROLLMENT;

SID
1
2
3
4
5
6

INTERSECT Example (Common IDs)

Conceptually:

SELECT SID FROM STUDENT

INTERSECT

SELECT SID FROM ENROLLMENT;

SID
1
2

Option 1(MySQL)

```
SELECT DISTINCT S.SID
FROM STUDENT S
INNER JOIN ENROLLMENT E
ON S.SID = E.SID;
```

Option 2(MySQL)

SELECT SID

FROM STUDENT

WHERE SID IN (SELECT SID FROM ENROLLMENT);

MINUS / EXCEPT Example

Query: Students present in STUDENT but not in ENROLLMENT

Conceptually:

SELECT SID FROM STUDENT

EXCEPT

SELECT SID FROM ENROLLMENT;

SID
3
4
5

Option 1(MySQL)

```
SELECT SID
FROM STUDENT
WHERE SID NOT IN (SELECT SID FROM ENROLLMENT);
```

Option 2(MySQL)

```
SELECT S.SID
FROM STUDENT S
LEFT JOIN ENROLLMENT E
ON S.SID = E.SID
WHERE E.SID IS NULL;
```

Assignment Operator

Example: Update marks of Rohit to 75

UPDATE STUDENT

SET Marks = 75

WHERE Name = 'Rohit';

SID	Name	Marks
3	Rohit	75

SQL JOINS

Need	Use
Common records only	INNER JOIN
All from left table	LEFT JOIN
All from right table	RIGHT JOIN
Join condition uses =	Equi Join

SYNTAX: JOIN ON 2 RELATION

```
SELECT column_list
FROM Table1
INNER JOIN Table2
ON Table1.common_column =
Table2.common_column;
```

SYNTAX: join ON 3 relations

```
SELECT column_list
FROM Table1
JOIN Table2 ON condition
JOIN Table3 ON condition;
```

SQL JOIN Examples

Tables:

- EMPLOYEE(EmpID, EmpName, DeptID, Salary)
- DEPARTMENT(DeptID, DeptName)
- PROJECT(ProjectID, ProjectName, DeptID)
- WORKS_ON(EmpID, ProjectID)

Example 1: List employees with their department names.

```
SELECT E.EmpName, D.DeptName
FROM EMPLOYEE E
JOIN DEPARTMENT D
ON E.DeptID = D.DeptID;
```

Example 2: Department Wise Employee List

```
SELECT D.DeptName, E.EmpName
FROM EMPLOYEE E
JOIN DEPARTMENT D
ON E.DeptID = D.DeptID
ORDER BY D.DeptName;
```

Example 3: Projects Under IT Department

```
SELECT P.ProjectName
FROM PROJECT P
JOIN DEPARTMENT D
ON P.DeptID = D.DeptID
WHERE D.DeptName = 'IT';
```

Example 4: Employees Not Assigned Any Project (LEFT JOIN)

```
SELECT E.EmpName
FROM EMPLOYEE E
LEFT JOIN WORKS_ON W
ON E.EmpID = W.EmpID
WHERE W.ProjectID IS NULL;
```

Example 5: Employees Working on More Than One Project

```
SELECT E.EmpName
FROM EMPLOYEE E
JOIN WORKS_ON W ON E.EmpID = W.EmpID
GROUP BY E.EmpName
HAVING COUNT(W.ProjectID) > 1;
```